MILITARY SPECIFICATION

CABLES, POWER, ELECTRICAL, FLAT, THREE-CONDUCTOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flat, three-conductor power cables. Each conductor is individually covered and laid parallel to each other under a jacket of chloroprene rubber. (See 6.1.)

1.2 Part number. The part number is M55483-1 or M55483-2 (see 6.2 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-3930 - Insulating and Jacketing Compounds, Electrical (for Cables, Cords, and Wires).

MIL-C-12000 - Cable, Cord, and Wire, Electric; Packaging of.

STANDARDS

FEDERAL

FED-STD-228 - Cable and Wire, Insulated; Methods of Testing.

MILITARY

MIL-STD-104 - Limits for Electrical Insulation Color.

MIL-STD-105 - *Sampling Procedures and Tables for Inspection* by Attributes.

MIL-STD-169 - Extreme-Temperature Cycle.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications - Electronics Command, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B33 - Standard Specification for Tinned Soft or Annealed Copper Wire for Electrical Purposes.

ASTM B172 - Rope-Lay-Stranded Copper Conductors Having Bunch-Stranded Members, for Electrical Conductors.

ASTM B189 - Standard Specification for Lead-Coated and Lead-Alloy-Coated Soft Copper Wire for Electrical Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Insulation. The insulation of the individual conductors for M55483-1 shall be type IR or type IS-L in accordance with MIL-I-3930 using nonstaining anti-oxidants and anti-ozidants. The insulation of individual conductors for M55483-2 shall be type IH in accordance with MIL-I-3930.

3.1.2 Cable tapes. If tapes are used over the individual insulated conductors, they shall be rubber or rubber like compound-filled fabric cable tapes.

3.1.2.1 Material. Compound-filled tape shall be made from cloth frictioned with an insulating compound. The thickness shall be not less than 0.010 inch. Color shall be black, white (or natural), and green (see 3.2.1).

3.1.2.2 Application. Tape shall be applied helically and overlap not less than 10 percent of its width. When applied directly over the insulation, it may be applied either before or after vulcanization.

3.1.3 Braids. If braids are used over the individual conductors, they shall be of the close weave type. Color shall be black, white (or natural), and green (see 3.2.1) corresponding to the insulation color (see 3.2.2.1).

3.1.4 Jacket. Prior to jacketing, it is acceptable for the cabled conductors to be reinforced with an open mesh dry braid to facilitate the jacketing process. The braid shall not add to the overall dimensions of the cable. The cable shall be jacketed with type JN-L chloroprene rubber in accordance with MIL-I-3930. The jacket shall be vulcanized and free stripping.

3.2 Construction. The cable shall be constructed in accordance with the following requirements (see 4.6):

3.2.1 Conductors. The cable shall consist of three conductors No. 6 AWG in size. Each conductor shall conform to ASTM B172, class K.

3.2.1.1 Conductor strands. Each strand of wire shall be No. 30 AWG, annealed and coated copper wire. The tensile, electrical, and coating properties shall comply with ASTM B33 or ASTM B189. The coating shall be continuous tin or tin lead alloy.

3.2.2 Insulation thickness. The average thickness of the insulation shall be not less than 0.063 inch. The minimum thickness at any point shall be not less than 90 percent of the specified thickness.

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3.2.2.1 Color. The first conductor insulation color shall be white; the center conductor insulation shall be green; and the other outer conductor insulation shall be black. Colors shall be in accordance with MIL-STD-104.

3.2.3 Jacket thickness. The average thickness of the jacket shall be not less than 0.078 inch. The minimum thickness shall be not less than 90 percent of the specified thickness.

3.2.4 Outside diameter. Outside diameters of the cable shall be 0.522 ±0.030 inch minor diameter and 1.212 ±0.040 inches major diameter.

3.3 Manufacturing repairs. Manufacturing repairs shall conform to the following requirements:

3.3.1 Conductor strand splices. When splices are required in the individual strands, they shall be of the brazed butt type. The brazed joint shall be the same diameter as the original wire. Dressing of splices or removal of burrs is acceptable to meet this requirement. Multiple strand splices may be made in the stranding operation by joining the individual strands provided that the individual strand splices are spaced a minimum of 1 foot apart. No strands shall be kinked and the smoothness of lay shall not be altered. There shall be no splices or joints in the conductor as a whole.

3.3.2 Insulation repairs. The insulation shall be cut and removed from any faulty section. The exposed conductor shall be covered with thin layers of rubber tape, tightly applied, overlapping the original insulation at least one-quarter of an inch on each side and vulcanized in a suitable mold. The repaired cable shall be capable of meeting the requirements of this specification.

3.3.3 Jacket repairs. If repairs to the jacket are necessary, the materials and methods to be used shall be subject to the contracting officer's approval.

3.4 Electrical requirements.

3.4.1 Dielectric strength. When tested as specified in 4.6.2, the cable shall show no visual evidence that rupture of the insulation has occurred.

3.4.2 Insulation resistance. When measured as specified in 4.6.3, the insulation resistance shall be not less than the figure obtained from the following formula:

$$R = K \log \frac{D}{d}$$

Where:

R = Insulation resistance in megohms - 1,000 feet
K = 2,000
d = Average diameter under insulation
D = d + 2C
C = Specified nominal insulation thickness (see 3.2.2)

In using the above formula, all dimensions shall be expressed in the same units.

3.5 Environmental requirements.

3.5.1 Low temperature flexibility. When tested as specified in 4.6.4, the finished cable shall exhibit no cracks or any other signs of damage when examined under a glass having a magnification of 3 diameters. First and last turns shall be disregarded.

3.5.2 Extreme temperature cycling. During the test specified in 4.6.5, the finished cable shall meet the electrical requirements of 3.4 and shall show no evidence of deterioration.

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3.6 Marking. The cable shall be marked on the outside with an inked marking giving name of manufacturer, part number, and the year of manufacture, with the spacing of 3 to 5 letters being allowed between them. The marking shall be on the flat side and shall be horizontal to the lengthwise axis of the cable. All letters and numbers shall be .188 ±.001-inch high. Inked marking shall be clear and legible contrasting with the basic color of the jacket. The marking shall be repeated at intervals along the cable not to exceed 1 foot. The outer surface of the cable shall be smooth and free from raised or indented markings.

3.7 Workmanship. The cable shall be manufactured in such a manner as to be of uniform quality and free from any defects that would affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.3).

b. Quality conformance inspection (see 4.5).

4.3 Materials inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Requirement paragraph
Insulation	3.1.1
Jacket	3.1.4
Conductor strands	3.2.1.1

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the conditioning requirements specified in FED-STD-228.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Unit of product. Unless otherwise specified in the contract or purchase order (see 6.2), the unit of product for purposes of sampling shall be each continuous length of cable contained on a reel, spool, or coil.

4.5.1.2 Lot size. The lot size shall be the number of units of product as determined in 4.5.1.1 offered for inspection at one time for group A inspection, and on a monthly production basis for group B inspection.

4.5.1.3 Sample. The sample shall consist of that number of sample units required by the sampling plan for the lot size as determined in 4.5.1.2.

4.5.1.4 Selection of sample. When applicable, the number of sample units required for inspection shall be chosen by selecting one coil, spool, or reel for each sample unit required, after which the coil, spool, or reel shall be treated as the sample unit for purposes of inspection. Otherwise the sample units shall be selected at random without regard to their quality except that the samples selected at the start of the contract shall be selected from the first units produced.

4.5.1.5 Specimen. A specimen is an individual piece of cable taken from a sample unit.

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4.5.1.6 Sampling plan. For groups A and B inspections, sampling shall be in accordance with MIL-STD-105, special inspection level S-2, and an acceptable quality level of 1.0 for major defects and 4.0 for minor defects (see tables I, II, and III). All electrical defects shall be considered major defects.

4.5.2 Group A inspection. This inspection shall conform to table I and the normal inspection procedures of MIL-STD-105. Group A inspection shall be performed in any order, unless otherwise specified in the contract or purchase order.

4.5.2.1 Reworked lots. Lots which have failed 4.5.2 shall be reworked and reinspected to the tightened inspection procedures of MIL-STD-105.

4.5.2.2 Rejected lots. Lots which have failed 4.5.2.1 shall be discarded.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual and mechanical inspection- - -	3.1, 3.2, 3.3, 3.6, and 3.7	4.6.1
Electrical:		
Dielectric strength- - - - - - - - -	3.4.1	4.6.2
Insulaton resistance - - - - - - - -	3.4.2	4.6.3

4.5.3 Group B inspection. This inspection shall consist of the tests specified in table II and shall be performed every month on sample units selected from inspection lots that have been subjected to and have met group A inspection.

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Low temperature flexibility- - - - -	3.5.1	4.6.4
Extreme temperature cycling- - - - -	3.5.2	4.6.5

4.5.3.1 Noncompliance. The contractor shall immediately report, in writing, each group B failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the processes and materials, as applicable, and (2) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract or purchase order (see 6.2) through the quality assurance representative. After corrective action has been taken, additional sample units shall be subjected to group B inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and group A inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group B reinspection results have shown that the corrective action was effective.

4.5.3.2 Rejected lots. Lots which have failed 4.5.3.1 shall be discarded.

4.5.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-12000.

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4.6 Inspection and examination.

4.6.1 Visual and mechanical inspection. The finished cable shall be examined for the defects listed in table III and shall meet the requirements of 3.1, 3.2, 3.3, 3.6, and 3.7.

4.6.2 Dielectric strength. The finished cable shall be tested in accordance with method 6111 of FED-STD-228 for compliance with 3.4.1. The source of potential shall be a generator or transformer capable of delivering at least 1 kilovolt ampere at normal load. A potential of 3500, +100, -0 volts rms shall be maintained for 5 minutes minimum between each conductor and ground and between both conductors while the cable is immersed in water.

TABLE III. Classification of visual and mechanical defects.

Classification	Defect
Major	Cable tapes not correct (3.1.2). Braid not correct (3.1.3). Jacket not free stripping (3.1.4). Incorrect number of strands (3.2.1). Strands not correct size (3.2.1.1). Jacket not correct thickness (3.2.3). Outside cable diameter not correct (3.2.4). Repairs not meeting this specification (3.3).
Minor	Insulation not centered (3.2.2). Incorrect insulation thickness (3.2.2). Identification marking not correct (3.6).

4.6.3 Insulation resistance. The finished cable shall be tested in accordance with method 6031 of FED-STD-228 for compliance with 3.4.2. The insulation resistance shall be computed from the galvanometer deflection obtained after an electrification of 1 minute minimum with the negative pole of the source of potential connected to the conductor. The potential shall be 100 volts dc minimum. The insulation resistance may be measured prior to 1 minute if the galvanometer ceases fluctuating and reading indicates that a steady or increasing resistance value has been obtained.

4.6.4 Low temperature flexibility. Specimens in a nonflexed position shall be placed in a cold chamber maintained at a temperature of -55°C ±2°C for a minimum of 48 hours. At the end of this time and while still in the cold chamber, each specimen shall be wound around a mandrel having a diameter a maximum of four times the diameter of the cable, for five turns at a rate of one complete turn per minute. Upon removal from the cold chamber, the specimens shall be examined for compliance with the requirements of 3.5.1.

4.6.5 Extreme temperature cycling. The finished cable shall be subjected to the extreme temperature cycle shown in MIL-STD-169. Measurements shall be taken on steps 1, 3, 4, 5, 7, 8, 9, and 10 and shall meet the requirements of 3.5.2.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-12000.

6. NOTES

6.1 Intended use. The cable is intended for use as a portable cable for supplying electrical power.

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6.2 Ordering data. The contract or purchase order should specify the following:

a. Title, number, and date of this specification.

b. Military part number of cable (see 1.2).

c. Length of unit of product (see 4.5.1.1).

d. Responsible technical activity (see 4.5.3.1).

6.3 Superseding information. Part number M55483-1 supersedes type WT-26( )/U and is one way interchangeable with it. Part number M55483-2 is the same as M55483-1 except for the newer insulation material. Potting compounds required in cable assemblies may be different for the two part numbers due to adhesion characteristics.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - YD
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 17
DLA - ES, IS

Agent:
DLA - ES

Preparing activity:
Army - CR

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-55483 A CABLES, POWER, ELECTRICAL, FLAT, THREE-CONDUCTOR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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